

Suryansh Gupta

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SUMMARY

Machine Learning and Generative AI-focused developer with hands-on experience in data science, NLP, large-scale data processing, and LLM workflows. Experienced with Python, SQL, Scikit-learn, TensorFlow/Keras, Hugging Face Transformers, embeddings, and vector search. Currently building LLM-based applications and exploring parameter-efficient fine-tuning with PEFT, LoRA, and QLoRA.

TECHNICAL SKILLS

- **Programming:** Python, C++, SQL
- **ML/Data:** Scikit-learn, Pandas, NumPy, Feature Engineering, Time-Series, Clustering, PCA
- **Deep Learning:** TensorFlow, Keras, LSTM, RNNs, Transformers
- **NLP/GenAI:** Hugging Face Transformers, PEFT, LoRA, QLoRA, SentenceTransformers, Embeddings, FAISS, NLTK
- **Frameworks/Tools:** FastAPI, Streamlit, Git, REST APIs, LM Studio, Groq API, Vercel, Render
- **Currently Learning:** LangChain, LangGraph, LLM application development and deployment

EXPERIENCE

SmartED Innovations — Data Science Intern | Remote | Jan 2026 – Apr 2026

- Architected end-to-end automated data ingestion and preprocessing pipelines for **58M+ retail records**, implementing dynamic column mapping, time-series feature engineering, and robust cleaning workflows that reduced manual data-preparation time by **80%**.
- Applied **LightGBM** for large-scale sales forecasting and **K-Means/DBSCAN** for RFM-based customer segmentation, with emphasis on data quality and reliable analytical workflows.
- Built interactive **Streamlit/Plotly** dashboards for model and pipeline metrics, supporting local model training, automated batch exports, and stakeholder reporting.
- Developed a Retail Customer Clustering System processing **500k+ rows**, resolving **135k+ missing values/outliers**, applying RFM analysis and PCA, and optimizing clustering performance.
- Built a Retail Sales Forecasting System using the **M5 Walmart dataset**, including automated CSV column mapping, time-series feature engineering, and real-time metric visualization.

Teachnook — AI Intern | Remote | Aug 2024 – Sep 2024

- Designed and deployed a **Sentiment Analysis** pipeline using Python and NLTK for text ingestion, preprocessing, and multi-class emotion/sentiment classification.
- Implemented tokenization, stemming, stop-word removal, and Scikit-learn classifiers for automated text labeling.
- Handled the application lifecycle from data collection and preprocessing through model evaluation and deployment.

SELECTED PROJECTS

CS-NLP Toolkit — In Progress — Building a reusable NLP/LLM toolkit around Hugging Face Transformers for computer-science text tasks, with experiments in pretrained-model inference and parameter-efficient fine-tuning using PEFT, LoRA, and QLoRA.

Local AI Companion — Developed a local JARVIS-style assistant using an LLM served through LM Studio, SentenceTransformers embeddings, FAISS vector search, and metadata-based memory; designed for persistent retrieval over previous conversations.

Rural India Translator — Built a translation application with a FastAPI backend and React/Tailwind frontend, experimenting with Hugging Face MarianMT models and Coqui TTS for speech output.

EDUCATION

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat, India — B.Tech. in Mechanical Engineering | CGPA: **8.09/10** | Graduated 2026

Relevant Coursework: Machine Learning, Data Structures & Algorithms, Probability & Statistics, Linear Algebra, Database Management Systems

CERTIFICATIONS

SmartED Certified Data Science (2026) | Teachnook Certified Artificial Intelligence (2024) | Generative AI Certified — NxtWave (2024)